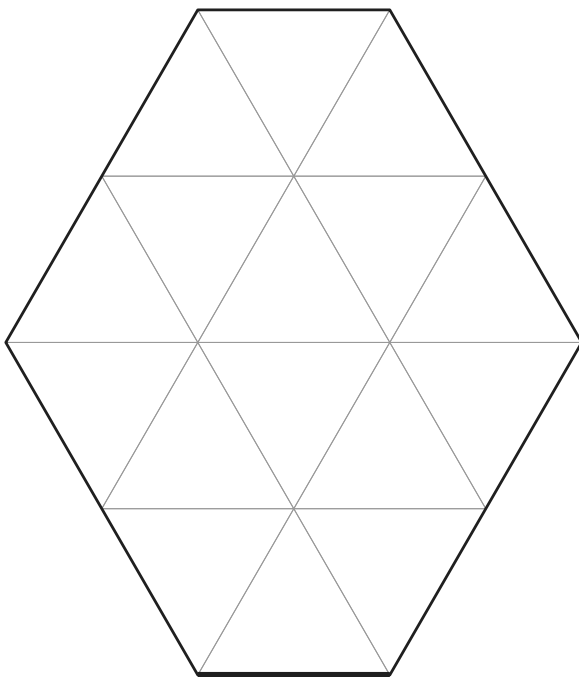
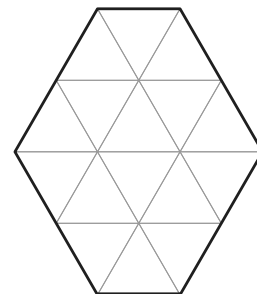
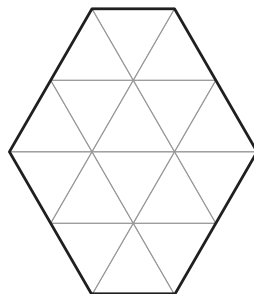
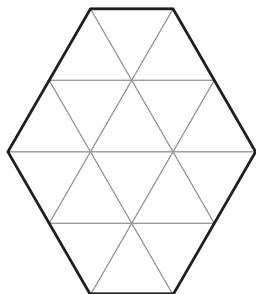
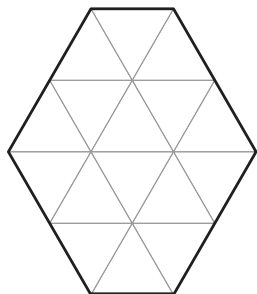
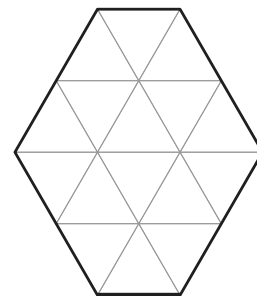
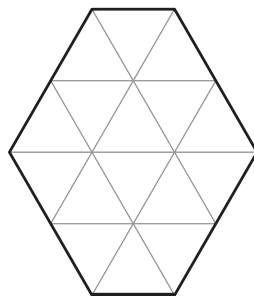
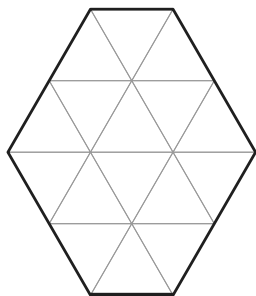
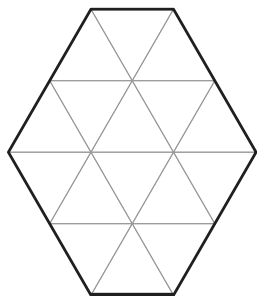


Welcome, tiling explorers! In this activity, you will discover the secret world of flips.

Problem 1: Fill the large board using only small blue rhombi. Find as many different tilings as you can. Keep the bold edge at the bottom. Record each tiling on a small board.

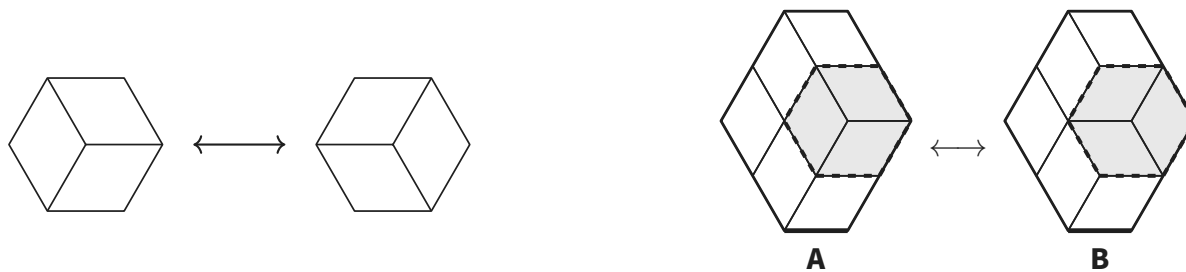


Record each blue rhombus with one line joining the centers of its two triangle spaces.

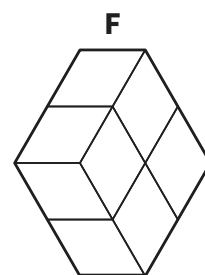
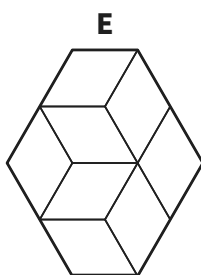
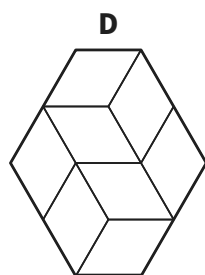
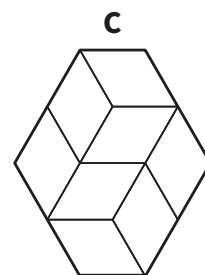
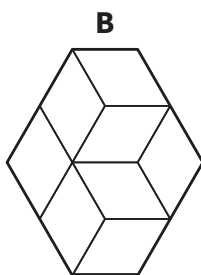
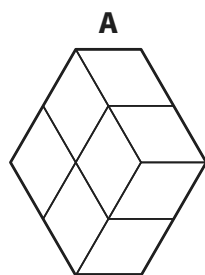


How many different tilings did you find? _____

Problem 2: A flip changes three blue rhombi inside a small hexagon to its other filling. The other rhombi stay in place. Try the flip shown from A to B, then flip back.



Make a map on these six cards. Draw a line between two cards whenever one flip changes one into the other. Check each line with your blocks.



Can you get from every card to every other card by a sequence of flips? Explain your thinking.

Problem 3: Find the shortest route from A to F. Write the cards you visit and count your flips. Explain your thinking.

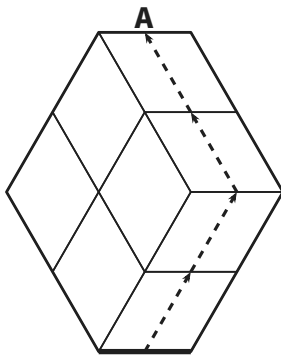
Route: A _____

Number of flips: _____

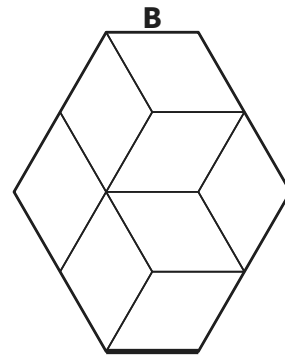
Problem 4: Can you leave A and return to A in exactly three flips? Five flips? Any odd number of flips? Explain what you notice.

Part 2: Ribbons, the Secret Code

Problem 5: Follow a ribbon from the middle of the bold bottom edge. Cross its rhombus to the middle of its opposite horizontal edge. Keep crossing upward until you reach the top. Follow the example on A, then draw the ribbon on B. Write L for each step to the left and R for each step to the right, reading from bottom to top.



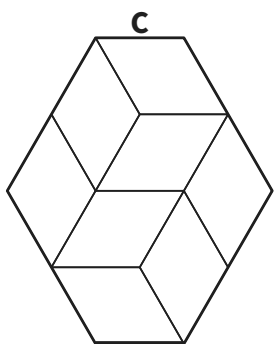
Code: _____



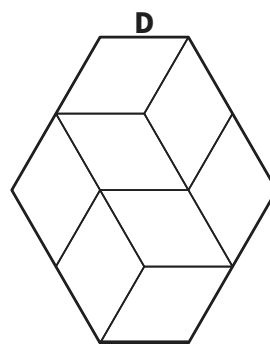
Code: _____

At any step, is there a choice of which rhombus to cross next? Could one tiling have two different ribbons starting at the bold edge?

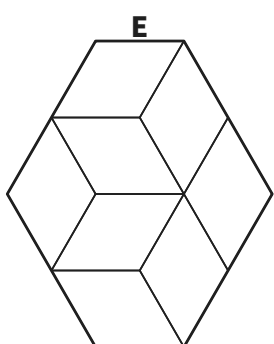
Problem 6: Draw the ribbon on each of these tilings. Write its code from bottom to top.



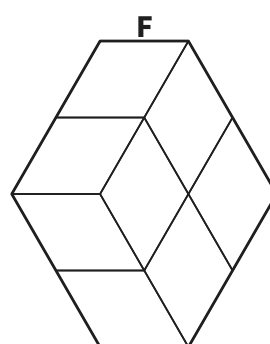
Code: _____



Code: _____



Code: _____



Code: _____

Problem 7: Compare the six ribbon codes from Problems 5 and 6. Follow these steps. (a) Count the L's in each code. (b) Count the R's in each code. (c) Add them up. (d) Look at how many edges the board has on its left and right sides, and use that to explain why.

Write every different order you can make with two L's and two R's. Match each order to a tiling card.

Code: _____

Card: _____

Code: _____

Card: _____

Code: _____

Card: _____

Code: _____

Card: _____

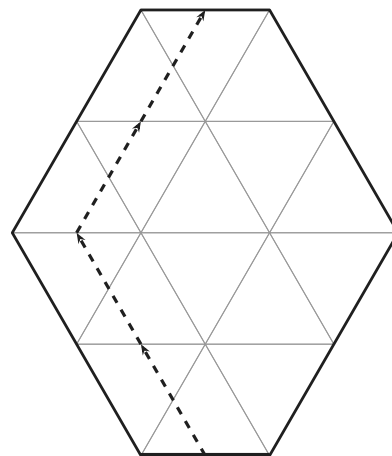
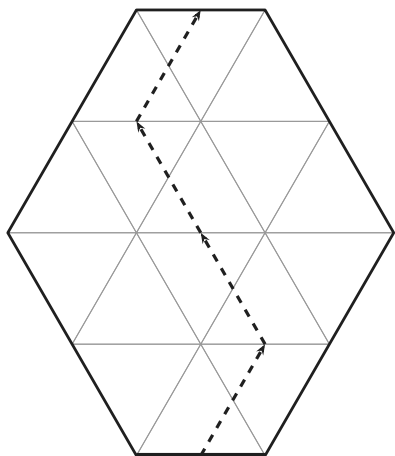
Code: _____

Card: _____

Code: _____

Card: _____

Problem 8: Rebuild the tiling from each ribbon. Draw the four blue rhombi crossed by the ribbon, then fill the remaining spaces with blue rhombi. Can either ribbon give you two different tilings?



Problem 9: Compare the ribbon codes before and after each flip on your map. What changes? Write three examples.

Cards	Code before	Code after
_____	_____	_____
_____	_____	_____
_____	_____	_____

Problem 10: Circle every neighboring RL or LR in your six codes. Can you make a flip at every circled pair? Are there any flips that do not come from one of these pairs? Check with the cards and blocks, then describe your rule.
